Department of Civil Engineering

Academic Year: 2024- 2025(Odd/~~Even~~ Semester)

Degree, Semester & Branch: B.E., I & CIVIL

Course Code & Title: GE3151 & Problem solving and python programming

Name of the Faculty member (s): Ms.M.Preethi Ram, AP/CSBS

Innovative Practice Description

- **Unit / Topic: Unit 2 / Values and types: int, float, Boolean**
- **Course Outcome: CO2**
- **Topic Learning Outcome: TLO4**
- **Activity Chosen: Flipped classroom**

  Justification: Flipping the classroom is a pedagogical approach that reverses the conventional teaching method. In this method, the traditional in-person teaching is restructured to cater to the unique learning requirements of individual students. Students take charge of their learning by independently studying course materials before attending class, such as reading materials and pre-recorded video lectures. This approach allows educators to reimagine in-class activities, which may include interactive problem-solving sessions, aimed at keeping students actively involved in the subject matter.

  Time Allotted for the Activity: 30 Minutes

- **Details of the Implementation:**
  - The learning materials were posted to the students in canvas two days before the topic discussion.
  - The students are instructed to study the material and take notes on the topic based on their understanding.
  - On the day of the activity the students were grouped into a team of 2 members and they are asked to discuss about the topic they learnt in their home through the study materials for the duration of 10 minutes.
  - Based on the study material they have gone through, each group prepare separately and by discussing with their team members they have clear idea of it.
  - After the discussion one member from each team had a chance to present and share the way of methods to the entire class as shown in Fig.1 & 2.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PSO3
CO2	2	2	2	1

- **PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PSO3**
	2	**2**	**2**	**1**
Justification for correlation	Students can able to develop python program to solve the various engineering problems by using basics mathematics, science and engineering principles	Students can able to understand the various python programming data types, operators and its precedence can be identified, formulated to solve the simple and complex engineering problems	Students can be able to provide solution to the real time problems can be solved by using necessary data types, operators in python.	Students can be able to create and demonstrate simple python program to solve the civil engineering problem.

• Images / Screenshot of the practice:

Fig 1: Presentation done by Prem Sundar T

Fig 2: Presentation done by Venkatesh

- **Reflective Critique:**
- **Feedback of practice from students and other stakeholders**:
  - ✓ Student felt good, since they can study at their own pace/time.
  - ✓ They felt that through such learning, they can explore more.
- **Benefit of the practice:**
  - ✓ More one-to-one time between teacher and student.
  - ✓ More collaboration time for students.
  - ✓ Students learn at their own pace.
  - ✓ Practical things – like missing class due to illness – become less problematic.
  - ✓ It encourages students to come to class prepared.

- **Challenges faced in implementation:**
    - ✓ Relies on student preparation – few students did not come prepared.
    - ✓ The depth of the subject can be dictated by the student themselves or the group the student is working with.
    - ✓ The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- https://www.teachthought.com/learning/the-definition-of-theflippedclassroom/#:~:text=A%0 flipped%20classroom % 20is%20a,the% 20students%20in dependently%20at%20home.
- https://en.wikipedia.org/wiki/Flipped_classroom
- youtube.com/watch?v=BCIxikOq73Q
- https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-whatwhy-and-how-to-implement-a-flipped-classroom-model

Signature of Faculty Member **HOD**